SECTION 2: MATHEMATICS PROFICIENCY
Time -- 60 minutes
60 Questions (101 - 160)

NAME: _____
SCHOOL: _____
SIMULATED EXAM - SET B

For each question, decide which is the best answer choice, and darken the corresponding oval on the answer sheet.

101. A bag contains 6 blue balls, 15 green balls, and a certain number of red balls. The bag is shaken before Andrew randomly chooses one ball from the bag. If the probability that Andrew doesn't pick a red ball is 0.7, how many red balls are there in the bag?
A. 3
B. 7
C. 9
D. 12

102. The slope of the line that passes through $(2, -6)$ and $(3, y)$ is 9. Find the value y.
A. -3
B. 0
C. 3
D. 5

103. Find the coordinates of the center and the length of the radius of the circle whose equation is $x^2 + y^2 - 12x + 8y + 47 = 0$.
A. $(6, -4)$, radius $= \sqrt{5}$
B. $(-6, 4)$, radius $= 5$
C. $(-6, 4)$, radius $= \sqrt{5}$
D. $(6, -4)$, radius $= 5$

104. Given that
$$\frac{9w + 7z}{3} = \frac{3w + 14z}{5}$$
Find the ratio w and z.
A. 7:36
B. 27:13
C. 31:25
D. 15:23

105. Solve for z: $log_{54} 3\sqrt[3]{2} = z$
A. $\frac{1}{3}$
B. $\frac{2}{3}$
C. 1
D. -1

106. Evaluate: $\log_{1/3}(\log_6 216)$
A. -2
B. -1
C. 0
D. 2

107. Given the logarithmic function $f(x) = \log_b x$ where $b > 0$ and $b \neq 1$, what is the value of $f(1)$?
A. b
B. 0
C. 1
D. -1

108. Determine the value of the reciprocal of the sum of the roots of the quadratic equation $(3x - 4)^2 - 5 = 0$.
A. $\frac{3}{8}$
B. $\frac{8}{3}$
C. $\frac{24}{11}$
D. $\frac{11}{9}$

109. What is the highest integral value of k for the quadratic equation $x^2 - 10x + k = 0$ to have real and unequal roots?
A. 24
B. 25
C. 26
D. 100

110. If 12, m, and 75 form a geometric sequence, which of the following could be a value for m.
A. -30
B. 15
C. -25
D. 36

111. Which of the following is an arithmetic sequence?
A. $-15, -13, -12, -10, \ldots$
B. $-4, 0, 4, 16, \ldots$
C. $9, 5, 1, -3, \ldots$
D. $0, 6, 36, 216, \ldots$

112. Find the domain of the function below.

$$f(x) = \frac{1}{\sqrt{-x}}$$

A. $x \geq 0$
B. $x > 0$
C. $x < 0$
D. $x \leq 0$

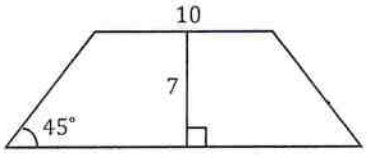
113. Let a and b be real numbers such that $a < b$. If c is any nonzero real number, then which of the following must be true?
A. $ac < bc$
B. $\frac{a}{c} < \frac{b}{c}$
C. $a + c < b + c$
D. $ac > bc$

114. Given that

$$x * y = \frac{x^y}{y^x}$$

Find the value of $2 * (2 * 2)$.
A. 1
B. 2
C. $\frac{1}{2}$
D. $\frac{1}{8}$

115. The lengths of the base and height of the given isosceles trapezoid are 10 cm and 7 cm respectively. What is the length of the other base?



A. 24 cm
B. 18 cm
C. 17 cm
D. 14 cm

116. What is the remainder when the polynomial $5x^4 - 2x^2 + 3x - 2$ is divided by $x + 3$?
A. -210
B. -103
C. 641
D. 376

117. Find the value of k such that when the polynomial $3x^4 + kx^3 + 5kx^2 + x - 2$ is divided by $x + 2$, the remainder is -28.
A. 8
B. 12
C. -6
D. -18

118. If $x+3$ is one of the factors of the polynomial $x^3+3x^2-5x+11k+7$, then find the value of k.
A. -15
B. -2
C. 7
D. 13

119. If $x+k$ is a factor of x^2-5x-6, which of the following could be the value of k?
A. 1
B. -1
C. 2
D. -2

120. Suppose $P(x)$ is a quadratic function with $P(1)=P(2)=0$ and $P(-1)=12$. Which of the following is not true?
A. $P(x)=2(x-1)(x-2)$
B. The roots of $P(x)=0$ are 1 and 2.
C. The $x-intercepts$ of the graph of $P(x)$ are 1 and 2.
D. $P(x)=x^2-3x+2$

121. Which of the following is(are) true about the graph of the rational function below?

$$f(x)=\frac{x+5}{x-2}$$

I. $x=2$ is a vertical asymptote.
II. The graph crosses the $y-axis$ at 5.
III. $y=1$ is a horizontal asymptote.

A. I only
B. I and II only
C. I and III only
D. I, II, and III

122. Suppose $f(x)$ is a linear function. If $f(5)=-8$ and $f(-1)=4$, find the value of $f(-7)$.
A. 16
B. 18
C. -24
D. -12

123. Solve for x:

$$(2^{4x})(3^{2x})=144$$

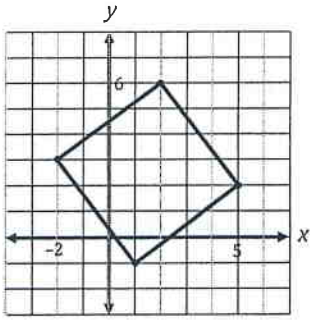
A. -1
B. 0
C. 1
D. 3

124. Simplify:

$$\frac{\sqrt[5]{25^{2x+2}\cdot 5^{x+1}}}{5}$$

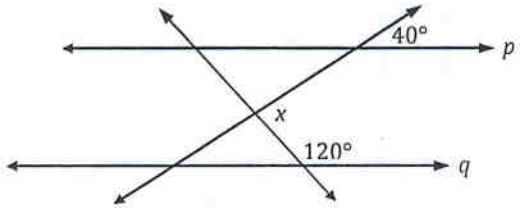
A. 5^{x+2}
B. 5^{x-2}
C. 5^{x-4}
D. 5^{x}

125. Find the area of the quadrilateral in the figure.



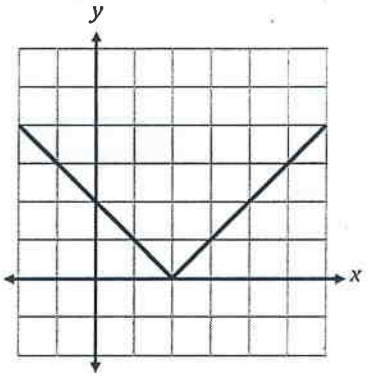
A. 12 *square units*
B. 16 *square units*
C. 25 *square units*
D. 28 *square units*

126. Refer to the figure. If $p//q$, then find the measure of angle x.



A. 160°
B. 100°
C. 90°
D. 40°

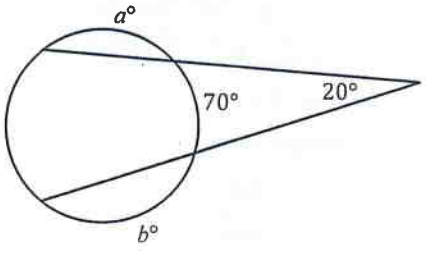
127. Given the graph below.



Which of the following could be its equation?

A. $y = |x - 2|$
B. $y = |x + 2|$
C. $y = |x| + 2$
D. $y = |x| - 2$

128. Refer to the figure. Find the value of $a + b$.



A. 180°
B. 160°
C. 110°
D. 90°

129. Solve for y:

$$\begin{cases} \frac{6}{x} + 2y = 2 \\ \frac{3}{x} - 3y = -7 \end{cases}$$

A. 2
B. 7
C. −3
D. −2

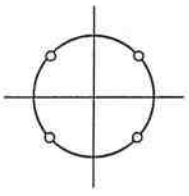
130. Simplify:

$$\frac{a^{-2} - b^{-2}}{a^{-1} - b^{-1}}$$

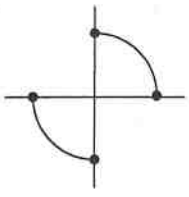
A. $\frac{a}{a-b}$
B. ab
C. $\frac{a+b}{ab}$
D. $\frac{a+b}{a-b}$

131. Which of the following graphs is a function?

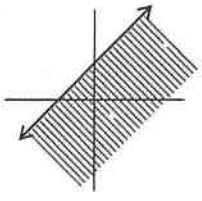
A.



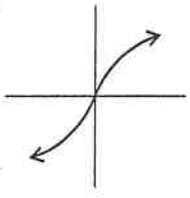
C.



B.



D.



132. Find the equation of the line that passes through the point $(-2, 5)$ and parallel to the line $y + 2 = -\frac{1}{2}x$.

A. $x + 2y + 9 = 0$
B. $2x - y - 1 = 0$
C. $x + 2y - 8 = 0$
D. $2x - y + 9 = 0$

133. What number is in the middle of the solution set of the compound inequality below?

$$3(x+2)-2(x+4) \geq -6$$
and
$$5(2-x)+2x \geq 4$$

A. 1
B. 0
C. −3
D. −1

134. Express $\left(\frac{x-y}{x^2-y^2}\right)^{\frac{1}{2}}$ in simplest form.

A. $\frac{1}{x+y}$
B. $\sqrt{x-y}$
C. $\frac{\sqrt{x+y}}{x+y}$
D. $\frac{\sqrt{x-y}}{x-y}$

135. Which of the following shows the Associative Property of Addition?

A. $[(-2)+(-3)]+(-5) = -5+[(-2)+(-3)]$
B. $[(-2)+(-3)]+(-5) = (-2)+[(-3)+(-5)]$
C. $[(-2)+(-3)]+0 = (-2)+(-3)$
D. $(-2)+(-3) = -5$

136. If $2x-3y=4$, express x^2-2x+3 in terms of y.

A. $\frac{9y^2+12y+16}{4}$
B. $\frac{9y^2+12y+12}{4}$
C. $\frac{9y^2-24y+16}{4}$
D. $9y^2+12y+12$

137. Given the system below.

$$3x+5y=30$$
$$kx+2y=12$$

Find the value of k so that the lines will not be unique.

A. 2
B. −2
C. $\frac{1}{2}$
D. $\frac{6}{5}$

138. If a and b are real numbers, then which of the following is not always true?

A. $|a+b| \geq a+b$
B. $|a-b| \geq a+b$
C. $|ab| \geq ab$
D. $|ab| \geq -ab$

139. The graph of a quadratic function is a

A. parabola that always intersect the y-axis once.
B. parabola that intersect the y-axis at least once.
C. parabola that intersect the y-axis at most once.
D. parabola that does not intersect the y-axis.

140. Which of the following is negative?

A. $\sin\frac{7\pi}{9}$
B. $\cos\frac{13\pi}{7}$
C. $\tan\frac{9\pi}{8}$
D. $\cot\frac{5\pi}{9}$

141. If $\tan\theta = \frac{4}{5}$ and $\cos\theta < 0$, then find the exact value of $\sin\theta$.

A. $\frac{4}{3}$

B. $-\frac{4}{3}$

C. $-\frac{4\sqrt{41}}{41}$

D. $\frac{4\sqrt{41}}{41}$

142. The terminal side of an angle θ in standard position passes through the point $(4, -3)$. Find the exact value of $\cos\theta$.

A. $\frac{3}{5}$

B. $\frac{4}{5}$

C. $-\frac{4}{5}$

D. $-\frac{3}{5}$

143. Which of the following is not equal to one?

A. $\frac{1}{\csc^2\theta} + \cos^2\theta$

B. $(\cos\theta + \sin\theta)^2$

C. $\sec^3\theta \cdot \cos^3\theta$

D. $\sec^2\theta - \tan^2\theta$

144. The graph of a quadratic function is a parabola that opens upward or downward. Given the quadratic function $y = x^2 + x - 15$, which point on the its graph has the same ordinate as the point $(3, -3)$?

A. $(-5, -3)$

B. $(-4, -3)$

C. $(1, -3)$

D. $(3, -4)$

145. Where does the graph of the quadratic function $f(x) = x^2 - 5x - 6$ crosses the $x - axis$?

A. at -1 only

B. at 2 only

C. at -1 and 6

D. at 2 and 3

146. Using the letters in the English alphabet, a four-letter code will be made. The first and third letters must be X or Y, but must not be the same simultaneously. The second and the fourth letters can be any letter from the English alphabet, and repetition is allowed. How many such codes can be made?

A. 26^2

B. $2 \cdot 26^2$

C. $26 \cdot 25$

D. $26 \cdot 25 \cdot 2$

147. Which of the following is equal to $\cos 60°$?

A. $\sin\frac{5\pi}{6}$

B. $\cos\frac{2\pi}{3}$

C. $\sec\frac{\pi}{6}$

D. $\csc\frac{\pi}{3}$

148. Which of the following show(s) Closure Property of Multiplication?

I. $0 + (-3)(5) = 0 + (-15)$

II. $[(-3)(5)](1) = (-3)(5)$

III. $(-3)(5) = -15$

A. III only

B. I and III only

C. II and III only

D. II only

149. Two congruent circles, A and B, with radii 15 intersect at exactly one point. If X lies on circle A and Y lies on circle B, what is the maximum length of line segment XY?
A. 15
B. 30
C. 45
D. 60

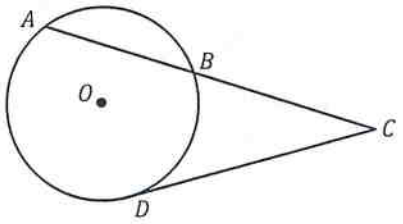
150. What is the area of a square inscribed in a circle whose diameter is 10 units long?
A. 25
B. 50
C. 75
D. $25\sqrt{2}$

151. A circle circumscribed a triangle whose sides are 18, 80, and 82. Find the length of the radius of the circle.
A. 9
B. 18
C. 40
D. 41

152. Find the $y-intercept$ of the graph of the line that is perpendicular to the line $3x = 2y + 9$ passes through the point $(-15, 8)$.
A. -5
B. -2
C. 2
D. 5

153. Which of the following ordered pairs is contained in the line whose slope is $\frac{2}{5}$ and crosses the $y-axis$ at 6?
A. $(-2, -3)$
B. $(5, -1)$
C. $(-15, 0)$
D. $(10, -5)$

154. In the figure, a secant line and a tangent line of circle O intersect at point C. If $AB = 12$ and $BC = 15$, find DC.



A. 10
B. 12
C. $6\sqrt{5}$
D. $9\sqrt{5}$

155. Given that $f(x) = 3x + 2$ and $g(x) = x^2 - 8$ where $f(a) = g(a)$. Which of the following could be the value of a?
A. -6
B. -2
C. 0
D. 4

156. If a, b, and c are positive real numbers and $a < b < c$, then which of the following must not be true?
A. $bc > ab$
B. $a + b < b + c$
C. $bc > ac$
D. $a + c < a + b$

157. What is the equation of the line that bisects the second and fourth quadrants of the Cartesian plane?
A. $y = -x + 1$
B. $y = 2x + 3$
C. $y = -x$
D. $y = x$

158. Simplify:

$$\left(\sqrt[5]{96x^6}\right)^{\frac{5}{4}}$$

A. $2x\sqrt[4]{6x^2}$

B. $2x\sqrt[5]{3x^2}$

C. $2x\sqrt[4]{6x}$

D. $2x\sqrt[5]{3x}$

159. z is directly proportional to x and inversely proportional to y. Which of the following could be an equation that shows the relationship between the variables?

A. $2 = \frac{y}{zx}$

B. $2 = \frac{zy}{x}$

C. $2 = \frac{xy}{z}$

D. $2 = \frac{z}{xy}$

160. Simplify the expression: $\frac{\sqrt{24x^5y^3}}{\sqrt{54xy^{11}}}$

A. $\frac{4x^2}{9y^4}$

B. $\frac{2x^2\sqrt{6x}}{3y^4}$

C. $\frac{2x^2\sqrt{xy}}{3y^4}$

D. $\frac{2x^2}{3y^4}$